Cost and scaling of cosine-filter ground-state preparation from a matrix-product state: gap, overlap and target in the J_1–J_2 chain, with a comparison to QETU

Authors to be added
(Dated: November 3, 2026)

A common route to a ground state on a quantum computer compiles a classical trial state and removes its residual excited weight with a quantum filter. We benchmark the cost of this route on the spin-$\frac{1}{2}$ J_1–J_2 chain, with a matrix-product-state (MPS) trial and the one-ancilla cosine filter of Stetcu, Baroni and Carlson (SBC), and compare plans certified by an a-priori bound with exact simulation of the Trotterized circuits on up to 16 qubits. A controlled study that holds the gap (0.30–0.55) and the overlap fixed separately at $N = 8$–14 shows that the gap sets the circuit depth and the overlap the repetitions, and gives fitted cost laws for both. The certificate's overhead is almost all in its Trotter bound: the planned filter is within 2.0× of the length simulation needs, but the bounded step is 60–75 times finer than needed, and the certified cost is a median 62× the measured one. Certified plans extend to $N = 48$. Against a Chebyshev (QETU) filter under identical accounting, SBC is cheaper measured at $\epsilon \approx 10^{-2}$, ties at 10^{-3} and loses from 10^{-4}, and certified it is not cheaper. The infidelity bound held in all 96 simulated circuits, all at $N \leq 14$ with exact inputs; for larger systems it is conditional on DMRG estimates of the gap and overlap, and a 10–25% gap overestimate breaks the optimized and Chebyshev plans. Measured against compiling a better MPS, the filter wins on chains only as a last-mile polisher (18 of 63 targets) but at 27 of 28 targets on small cylinders, where a snake MPS struggles; with certified plans it loses wherever an MPS reaches the target.

I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional systems the natural classical engine is DMRG [3, 4], which yields the ground-state energy E_0, the gap and a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [5]. The quantum step can be phase estimation [6], a polynomial filter implemented by quantum singular-value transformation [7–9], or a product of cosine filters implemented with a single ancilla [1, 13]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [1], needs little beyond time evolution: each pulse is one controlled time evolution between ancilla rotations and a measurement, the filter is designed classically, and the success probability is known in advance.

This paper is a cost benchmark of that workflow on the open J_1–J_2 Heisenberg chain. We ask what the filter costs, in two-qubit gates, arbitrary rotations, evolution time and repetitions, both as certified by an a-priori bound and as measured by exact simulation of the Trotterized circuit; what sets that cost; and how it compares with a Chebyshev polynomial filter (QETU) [9] and with simply compiling a better MPS, under identical accounting. The spectral gap Δ and the trial overlap γ normally change together with system size N and coupling. We decouple them: the gap is held fixed across N by tuning J_2, and the overlap by a one-parameter rotation of the trial circuit, so each can be varied while the other stays put.

Relation to prior work. The filter is SBC's [1]; it shares its circuit with the rodeo algorithm [13, 20], which draws the times at random. Our closed-form "dyadic" schedule, a halving ladder of times, is the one SBC and Rule *et al.* use for quantum-number projection [1, 21], and for energy it is a finite form of the rodeo super-iteration of Cohen and Oh [22], recently generalized [23]. The Trotter bounds are those of Childs *et al.* [2], the certificate follows the standard argument for polynomial filters [7, 9], and the scaling of filter depth with $1/\Delta$ and of repetitions with $1/\gamma$ is the known complexity of filter methods [6, 7]. QETU has been costed with Trotterized evolution for lattice gauge theories [24] and run on small Hubbard models [25, 26]. Resource studies that compile a classical trial and then refine it exist for quantum chemistry [11, 12, 27], and Watanabe *et al.* [28] stop an MPS encoder early and finish with probabilistic imaginary-time evolution on Heisenberg chains. What is new here is (i) a controlled separation of gap and overlap at fixed size, which turns the known scaling into measured constants and exponents, (ii) a measured, not only bounded, account of every resource and its factorization into filter length, Trotter step and gates per step, which shows where the certificate's overhead sits, and (iii) a like-for-like measured comparison of cosine and Chebyshev filters, including the Chebyshev filter's query length, and (iv) a measured benchmark of filtering a cheap MPS against compiling a better one, on chains and small cylinders. The method is not new; the contribution is the benchmark. The code is the `sbpipe` package, available with all data at `https://github.com/aaronseymour7/FilterPipeline`; every number below is produced by the scripts named in Appendix D.

Section II describes the pipeline and the certificate, Sec. III the model and the validation, Sec. IV the controlled gap/overlap study and its cost laws, Sec. V certi-

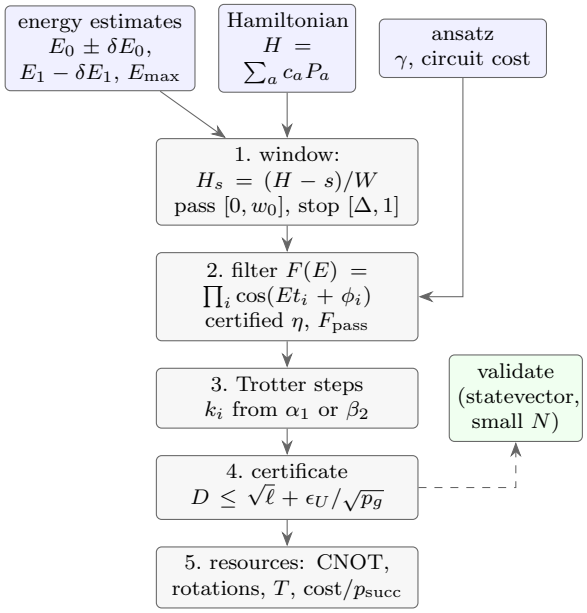


FIG. 1. The pipeline. Blue: inputs, which any classical method can supply. Grey: the stages of `refine()`. Green: the optional statevector check used in Sec. III A.

fied resources to $N = 48$ and their sensitivity to wrong inputs, Sec. VI the benchmark against the MPS alone, Sec. VII the comparison with QETU, and Sec. VIII the gap between measured and certified cost.

II. THE SBC FILTER AND THE RESOURCE PIPELINE

A. Inputs and the spectral window

The pipeline takes a Pauli-sum Hamiltonian $H = \sum_a c_a P_a$, an ansatz, and energy estimates with error bars,

$$\begin{aligned} E_0 &\in [E_0^{\text{est}} - \delta E_0,\ E_0^{\text{est}} + \delta E_0], \\ E_1 &\ge E_1^{\text{est}} - \delta E_1, \qquad E_{\text{top}} \le E_{\max}, \end{aligned} \tag{1}$$

where E_1 is the lowest level above the ground space that the trial state can overlap with; for trial states that are not symmetry-adapted, as here, this is the global first excited level. With an overestimated E_1 a real excitation can sit in the filter's transition region and the bound says nothing (Sec. V A). If no $E_{\max}$ is given, the rigorous Weyl bound $E_{\max} = \sum_g \lambda_{\max}(h_g)$ over term groups h_g is used. With $s = E_0^{\text{est}} - \delta E_0$ and $W = E_{\max} - s$, the scaled Hamiltonian $H_s = (H - s)/W$ has its ground energy in the *pass band* $[0, w_0]$, $w_0 = 2\delta E_0/W$, and every other relevant level in the *stop band* $[\Delta, 1]$, $\Delta = (E_1^{\text{est}} - \delta E_1 - s)/W$. The ansatz enters through γ, a lower estimate of its ground-space weight, and the gate cost of its circuit.

B. The filter and three schedules

One SBC pulse [1] applies $\mathsf{H}_a\, e^{-itH_s \otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; its ancilla-$|0\rangle$ block is $\cos(H_s t + \phi)$. Keeping outcome 0 after m pulses applies

$$F(H_s) = \prod_{i=1}^{m} \cos(H_s t_i + \phi_i), \qquad T = \sum_i |t_i|. \tag{2}$$

A filter is characterized by its pass-band floor $F_{\text{pass}} \le \min_{[0,w_0]} |F|$ and suppression ratio $\eta = \sup_{[\Delta,1]} |F|/F_{\text{pass}}$, both certified rather than sampled (by interval branch-and-bound using $|F'| \le T$, $|F''| \le T^2$, or in closed form). For a target leakage ϵ_ℓ the design target is $\eta^2 \le \gamma\epsilon_\ell/[(1-\epsilon_\ell)(1-\gamma)]$ (Lemma 1). Three schedules are provided.

Uniform: m pulses $t = \pi/(1+\Delta)$, $\phi = 0$. On the stop band $Et \in [\theta, \pi - \theta]$, $\theta = \pi\Delta/(1+\Delta)$, so $|\cos Et| \le \cos\theta$ and $T \approx 2\ln(1/\eta)/(\pi\Delta^2)$, quadratic in the inverse gap; it is the baseline.

Dyadic: r repetitions of the halving ladder $t = \tau, 2\tau, \ldots, 2^{K-1}\tau$, $\tau = \pi/(1+\Delta)$, $\phi = 0$. One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1} \cos(2^k E\tau) = \frac{\sin(2^K E\tau)}{2^K \sin(E\tau)}, \tag{3}$$

exact for every E and equal to 1 at $E = 0$. On the stop band $\sin(E\tau) \ge \sin\theta$, so a block is at most $\eta_b = 1/(2^K \sin\theta)$ there; this uses the scaling H_s and the choice of τ, and does not hold outside $[\Delta, 1]$. With $r = \lceil \ln\eta / \ln\eta_b \rceil$ blocks and $2^K \sin\theta \approx e$,

$$T_{\text{dyadic}} \approx \frac{e\,\tau}{\sin\theta} \ln\frac{1}{\eta} \;\approx\; \frac{e}{\Delta} \ln\frac{1}{\eta}, \tag{4}$$

the optimal $O(\Delta^{-1}\log\eta^{-1})$ scaling [7, 22] with an explicit constant and no optimization. The pipeline searches K and r for the smallest T.

Optimized: times and phases that maximize $F(0)^2$ subject to the certified constraint on η at fixed (T, m), over a grid of T and $m \in \{4, 6, 8\}$. This is SBC's energy recipe [1] with a certified constraint; nonzero phases trade success probability for a shorter T.

For each schedule the leakage share $\epsilon_\ell/\epsilon \in \{0.5, 0.25, 0.1, 0.05\}$ is scanned and the plan with the fewest expected CNOTs per success is kept; an "auto" mode does the same over the dyadic and optimized schedules together.

C. Trotterization

Each controlled evolution $e^{-it_i H_s \otimes Z_a}$ is approximated by k_i first-order (Lie) or second-order (Strang) steps over the non-identity terms; the identity term becomes one exact ancilla R_z per pulse. With $B_a = \sum_{b>a} H_b$, the

bounds of Childs *et al.* [2, Props. 15–16] are

$$\|S_1(t) - e^{-itH}\| \le \tfrac{t^2}{2}\alpha_1, \quad \alpha_1 = \textstyle\sum_a \|[B_a, H_a]\|, \tag{5}$$

$$\|S_2(t) - e^{-itH}\| \le t^3\beta_2, \quad \beta_2 = \tfrac{1}{12}\textstyle\sum_a \|[B_a,[B_a,H_a]]\| + \tfrac{1}{24}\textstyle\sum_a \|[H_a,[H_a,B_a]]\|, \tag{6}$$

evaluated for the terms of H_s (equivalently $\alpha_1(H)/W^2$ and $\beta_2(H)/W^3$ with physical time t_i/W); tensoring with Z_a leaves them unchanged. Each commutator is a short Pauli sum, and its norm is computed exactly on its support (dense up to 8 qubits, Lanczos to 10^{-12} up to 12). Every pulse evolves under the same $H_s \otimes Z_a$, so the per-pulse bounds share one constant, and minimizing $\sum_i k_i$ at fixed total error gives $k_i \propto |t_i|$; we take $k_i = \lceil |t_i|/\delta t \rceil$ with the common step δt of that solution.

D. The certificate

Theorem 1. *Assume (1) and that the trial state has ground-space weight at least γ. Let the designed filter have certified η and F_{pass}, let ϵ_U bound the operator-norm error of the implementation (Trotter error $\sum_i \epsilon_i$ plus $R\delta$ for the R arbitrary rotations of the trial and filter circuits, each synthesized to accuracy δ), and let*

$$\ell = \frac{(1-\gamma)\eta^2}{\gamma + (1-\gamma)\eta^2}, \qquad p_g = \gamma F_{\text{pass}}^2. \tag{7}$$

Let $|g\rangle = \Pi_0|\psi\rangle / \|\Pi_0|\psi\rangle\|$ be the normalized projection of the trial state onto the ground space; $|g\rangle$ is pure also when the ground space is degenerate. For a noiseless implementation the state $|\tilde{\psi}_{\text{out}}\rangle$ prepared conditioned on the ancilla outcomes 0 is pure, and

$$D \le \sqrt{\ell} + \frac{\epsilon_U}{\sqrt{p_g}}, \qquad 1 - \mathrm{Tr}(\Pi_0\tilde{\rho}) \le 1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2 = D^2, \tag{8}$$

where D is the trace distance between $\tilde{\rho} = |\tilde{\psi}_{\text{out}}\rangle\langle\tilde{\psi}_{\text{out}}|$ and $|g\rangle$. The success probability is at least $(\sqrt{p_g} - \epsilon_U)^2$. Below, $1 - F$ denotes $1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$, the usual infidelity for the nondegenerate chains studied here.

The proof (Appendix A) combines a leakage lemma for the exact filter with the observation that a vector error ϵ_U on an unnormalized vector of norm at least $\sqrt{p_g}$ moves its direction by at most $\epsilon_U/\sqrt{p_g}$. The pipeline sets $\epsilon_U = (\sqrt{\epsilon} - \sqrt{\ell})\sqrt{p_g}$, 90% of it for the product formula and the rest for rotation synthesis. If the trial already satisfies $1 - \gamma \le \epsilon$, the empty filter is offered too.

E. Resource model

We assume all-to-all connectivity and Clifford+R_z gates. For a Heisenberg bond (i, j) the three commuting controlled terms form one exact block,

$$e^{-ict(X_iX_j + Y_iY_j + Z_iZ_j)\otimes Z_a} = V^\dagger e^{-ict(Z_iZ_a + Z_jZ_a - Z_iZ_jZ_a)} V, \qquad V = \mathsf{H}_i\,\mathrm{CNOT}_{ij}, \tag{9}$$

whose diagonal factor is a phase polynomial on three parities [31]: 6 CNOTs and 3 R_z, against 12 CNOTs for three separate Pauli rotations and 8 for qiskit's `PauliEvolutionGate` (checked in the test suite). The block is the same unitary as its three term rotations, so the Trotter error is that of the term-wise formula. Control by an anticommuting Pauli string, as in controlled-reversal rodeo circuits [10] and control-free QETU [9], is not available, because no Pauli product anticommutes with all of XX, YY and ZZ on a bond. k Strang steps cost $k(2B-2)+1$ blocks for B bonds; each pulse adds one ancilla R_z. An arbitrary rotation at accuracy δ costs $3\log_2(1/\delta)$ T gates [14]. The expected cost per success is the cost of one attempt (trial plus filter) divided by the certified success probability $(\sqrt{p_g} - \epsilon_U)^2 \ge (1 - \sqrt{\epsilon})^2 p_g$ (repeat-until-success; early abort is not modeled).

III. MODEL, INPUTS AND VALIDATION

We study the open spin-$\frac{1}{2}$ chain

$$H = J_1 \sum_{i=1}^{N-1} \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_{i=1}^{N-2} \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \tag{10}$$

$J_1 = 1$, at $J_2 = 0$ (Heisenberg), $J_2 = 0.2411$ (near the gapless-to-dimerized transition [16, 17]) and $J_2 = 0.4$ (dimerized), for $N = 8$ to 48; it has $B = N - 1$ bonds at $J_2 = 0$ and $B = 2N - 3$ otherwise. The front end runs two-site DMRG (quimb [18]) with bond dimensions up to 256; E_1 is the lowest triplet, from a penalty run. The error bars are the energy change between the last two bond dimensions, a convergence heuristic, not a bound. Both DMRG energies are variational upper bounds, so a residual error in E_1 biases the gap upward, the dangerous direction. The trial is the variationally optimal MPS at bond dimension $\chi \in \{1, 2, 4, 8\}$, and γ is its overlap with the reference MPS, an estimate rather than a lower bound. An MPS of bond dimension 2^b is a staircase of $N - b$ unitaries on $b+1$ qubits, each costed by the quantum Shannon decomposition bound [15]. For $N \le 14$, exact diagonalization (ED) reproduces E_0 to 2.3×10^{-11}, the first excited level to 2.4×10^{-11} (always the triplet) and the overlaps to 4.0×10^{-8} (Table I). Beyond $N = 14$ the certificate is conditional on the DMRG gap and overlap.

A. Validation of the certificate ($N \le 14$)

For $N \in \{8, 10, 12, 14\}$, the three couplings, $\epsilon \in \{10^{-2}, 10^{-3}\}$, three schedules and both orders, we plan a circuit for the $\chi = 2$ trial and, where feasible, simulate

TABLE I. DMRG inputs: ground energy, gap $E_1 - E_0$ and trial overlap γ of the optimal bond-dimension-χ MPS. Source: `results/inputs.json`.

N	J_2	E_0	gap	$\gamma_{\chi=1}$	γ_2	γ_4	γ_8
8	0	-3.374933	0.3927	0.134	0.775	0.999	1.0000
12	0	-5.142091	0.2809	0.060	0.591	0.994	1.0000
16	0	-6.911737	0.2193	0.028	0.475	0.984	0.9999
24	0	-10.453786	0.1530	0.006	0.328	0.954	0.9994
32	0	-13.997316	0.1177	0.001	0.234	0.913	0.9978
48	0	-21.085956	0.0809	0.000	0.125	0.817	0.9902
8	0.2411	-3.135001	0.3859	0.113	0.904	1.000	1.0000
12	0.2411	-4.738077	0.2720	0.046	0.806	1.000	1.0000
16	0.2411	-6.343316	0.2100	0.019	0.703	0.998	1.0000
24	0.2411	-9.556221	0.1443	0.003	0.517	0.989	0.9997
32	0.2411	-12.770432	0.1099	0.001	0.373	0.972	0.9989
48	0.2411	-19.200243	0.0744	0.000	0.191	0.920	0.9946
8	0.4	-3.025191	0.4037	0.036	0.976	1.000	1.0000
12	0.4	-4.545626	0.2956	0.003	0.945	1.000	1.0000
16	0.4	-6.066666	0.2371	0.000	0.911	0.999	1.0000
24	0.4	-9.109325	0.1748	0.000	0.839	0.995	0.9998
32	0.4	-12.152237	0.1418	0.000	0.768	0.990	0.9995
48	0.4	-18.238247	0.1075	0.000	0.640	0.976	0.9981

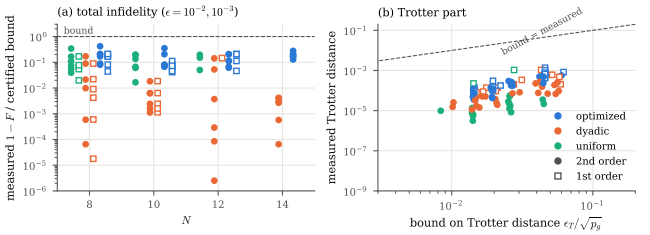


FIG. 2. All simulated plans. (a) Measured infidelity of the planned circuit divided by its certified bound; points below the dashed line respect the bound and are offset horizontally by schedule and order. (b) Measured distance between the Trotterized and the exact filtered state against its certified bound. Points below the dashed lines respect the bound. Filled circles: second order; open squares: first order.

it exactly against the ED ground state. Of 144 plans, 96 were simulated (the rest, mostly uniform or first-order plans with 10^4–10^8 steps, exceed our budget), and the certificate held in all 96 (Fig. 2).

The bound is conservative by a median factor 14 (range 2.4–4.0 $\times$ 10^5) in infidelity. The leakage term overestimates the exact filter's infidelity by a median 4.3×: 1.4 for the optimized schedule, which pushes $|F|$ to ηF_{pass} across the band, and 54 for the dyadic one, whose closed-form bound is reached only near the band edge. The Trotter term overestimates the measured Trotter distance by a median 185× (range 26–4.5 $\times$ 10^3): commutator bounds are worst-case over all states, while the filtered state lives in the low-energy subspace, where Trotter errors are much smaller [19, 29, 30]. The measured success probability equals p_g to two digits in every case.

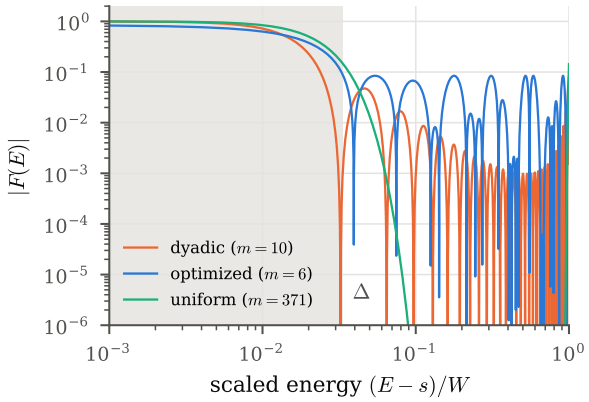


FIG. 3. $|F(E)|$ of the three schedules for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$, second order. Dotted lines: certified ηF_{pass}. Grey: the gap region. The horizontal axis is linear below Δ and logarithmic above.

Table II and Fig. 3 compare the schedules for one case. The uniform filter needs an order of magnitude more time, as its Δ^{-2} scaling predicts; the optimized filter is shortest but its phases lower p_g (0.58 against 0.81); second order cuts the step count by one to two orders of magnitude. For $N \le 12$ we also bisect a common scale factor on the certified k_i until the simulated infidelity just meets ϵ: the certified step count is 8.7–39× (median 21) this oracle value (Table III).

IV. GAP AND OVERLAP AS INDEPENDENT INPUTS

Along any natural family of chains the gap and the overlap move together: at fixed J_2 the gap closes with N, and at fixed χ the overlap decays with N. We therefore construct a family in which Δ and γ are set independently at every N.

Construction. For each $N \in \{8, 10, 12, 14\}$ and target gap $g = E_1 - E_0 \in \{0.30, 0.35, \ldots, 0.55\}$, J_2 is found by bisection on ED so that the singlet–triplet gap equals g to 10^{-9} (pairs without a solution below the dimerized regime are left out). For all tuned couplings the lowest $S^z = 0$ excitation coincides with the lowest triplet to 10^{-7}, so g is the true gap. The trial is the variational $\chi = 16$ MPS, prepared by its staircase circuit, followed by $R_y(\theta)$ on every even site; rotating one sublattice breaks the singlet symmetry and lowers γ smoothly, and θ is set by bisection to hit the target γ. The trial circuit is therefore the same along an overlap scan. Study A fixes $g \in \{0.40, 0.50\}$ and scans $\gamma \in \{0.03, 0.1, 0.3, 0.5, 0.7, 0.9\}$; study B fixes $\gamma \in \{0.1, 0.5\}$

TABLE II. Plans for $N = 12$, $J_2 = 0.2411$, $\chi = 2$ trial. $T_{\rm phys} = T/W$ in units of $1/J_1$; CNOT and T counts per attempt (filter) and expected per success (trial + filter, divided by the certified success probability $(\sqrt{p_g} - \epsilon_U)^2$). "rot.": arbitrary rotations per attempt (trial + filter), each synthesized to accuracy δ. "meas.": simulated infidelity of the planned circuit. Source: `results/validate.json`.

ϵ	schedule	order	m	$T_{\rm phys}$	p_g	steps	CNOT (filter)	CNOT/success	rot.	δ	T/success	meas. $1-F$
10^{-2}	uniform	1	437	164.2	0.81	4,136,205	5.2×10^{8}	7.2×10^{8}	2.6×10^{8}	2×10^{-11}	3.8×10^{10}	–
10^{-2}	uniform	2	371	139.4	0.81	17,437	4.2×10^{6}	5.5×10^{6}	2.1×10^{6}	1×10^{-9}	2.4×10^{8}	1.5×10^{-3}
10^{-2}	dyadic	1	10	23.3	0.81	77,308	9.7×10^{6}	1.4×10^{7}	4.9×10^{6}	1×10^{-9}	6.1×10^{8}	–
10^{-2}	dyadic	2	10	23.3	0.81	884	2.1×10^{5}	2.9×10^{5}	1.1×10^{5}	5×10^{-8}	1.1×10^{7}	8.5×10^{-7}
10^{-2}	optimized	1	6	8.7	0.58	13,384	1.7×10^{6}	3.2×10^{6}	8.4×10^{5}	5×10^{-9}	1.3×10^{8}	1.1×10^{-3}
10^{-2}	optimized	2	6	8.7	0.58	226	5.4×10^{4}	1.0×10^{5}	2.7×10^{4}	2×10^{-7}	3.5×10^{6}	1.1×10^{-3}
10^{-3}	uniform	1	745	279.9	0.81	27,817,555	3.5×10^{9}	4.5×10^{9}	1.8×10^{9}	1×10^{-12}	2.7×10^{11}	–
10^{-3}	uniform	2	658	247.2	0.81	55,930	1.3×10^{7}	1.7×10^{7}	6.7×10^{6}	2×10^{-10}	8.3×10^{8}	–
10^{-3}	dyadic	1	15	34.9	0.81	536,160	6.8×10^{7}	8.7×10^{7}	3.4×10^{7}	5×10^{-11}	4.5×10^{9}	–
10^{-3}	dyadic	2	15	34.9	0.81	2,835	6.8×10^{5}	8.8×10^{5}	3.4×10^{5}	5×10^{-9}	3.6×10^{7}	2.5×10^{-9}
10^{-3}	optimized	1	8	14.4	0.67	80,421	1.0×10^{7}	1.6×10^{7}	5.1×10^{6}	4×10^{-10}	7.4×10^{8}	–
10^{-3}	optimized	2	8	14.4	0.67	709	1.7×10^{5}	2.6×10^{5}	8.5×10^{4}	2×10^{-8}	1.0×10^{7}	6.2×10^{-5}

TABLE III. Certified over oracle step count (second order, $\chi = 2$). Columns: $N = 8, 10, 12$ for each J_2. Source: `results/oracle.json`.

		$J_2 = 0$			$J_2 = 0.2411$			$J_2 = 0.4$		
ϵ	schedule	8	10	12	8	10	12	8	10	12
1×10^{-2}	dyadic	13	22	30	24	24	34	18	15	28
1×10^{-2}	optimized	9	12	13	14	16	21	11	12	17
1×10^{-3}	dyadic	22	24	39	31	36	39	20	27	28
1×10^{-3}	optimized	12	17	21	24	22	27	14	16	17

and scans g; both at $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, 222 cases in all, each with the dyadic and optimized schedules and with QETU (Sec. VII).

Measured and certified costs. Measured: for the shortest filters that reach ϵ exactly (dyadic ladders, or SBC's optimized design for the true gap and overlap), the coarsest common Strang step at which the Trotterized circuit still reaches ϵ is found by bisection on the state vector. Optimized designs that barely pass the ground state ($p < \gamma/2$) are repeated over seeds (21 of 286 points). *Certified*: Theorem 1 with exact energies and the true γ. Costs are per shot and per success, for the filter alone unless stated.

A. The gap sets the circuit, the overlap sets the repetitions

Figures 4 and 5 show CNOTs, R_z rotations and evolution time per shot, and CNOTs per success, along both scans.

Per shot, the overlap enters only logarithmically. At fixed gap, lowering γ thirtyfold (0.9 to 0.03) raises the measured CNOTs per shot of the dyadic filter by only 1.2–3.0× and its evolution time by 1.5–3.0×. The filter must reach $\eta^2 = \gamma\epsilon/(1-\gamma)$, and a worse trial adds only $\ln 290 \approx 5.7$ to $L = \ln[(1-\gamma)/(\gamma\epsilon)]$, already 2–13 here. The dyadic filter grows in whole pulses, so its per-shot cost moves in steps.

Per shot, the gap enters as a power. At fixed overlap, closing the gap by 1.8× (0.55 to 0.30) raises the CNOTs per shot by 2.7–4.3× and the evolution time by 1.5–2.8×, about as $1/\Delta$. Both behaviors are the textbook $T = O(\Delta^{-1}\ln\eta^{-1})$ [6, 7]; the controlled study shows that the measured circuits follow it at fixed N with the inputs varied separately, and what the gates per shot and the Trotter step add.

Per success, the overlap returns as $1/\gamma$. Every run succeeds with probability $p \approx \gamma f(E_0)^2$, with $f(E_0) = 1$ for the dyadic filter, so the cost per success carries $1/\gamma$. The gap decides the depth of the coherent circuit; the overlap decides how often it has to be run.

R_z counts follow CNOT counts. Each bond block has 3 R_z and 6 CNOTs and each pulse adds one rotation, so $R_z/\mathrm{CNOT} = 0.50$ throughout; statements about CNOTs hold for the T count up to the synthesis factor.

B. Mechanism: filter length, Trotter step and gates per step

The per-shot cost factorizes as

$$\text{CNOTs per shot} = \frac{T}{\delta t} \times c_{\rm step}(N), \tag{11}$$

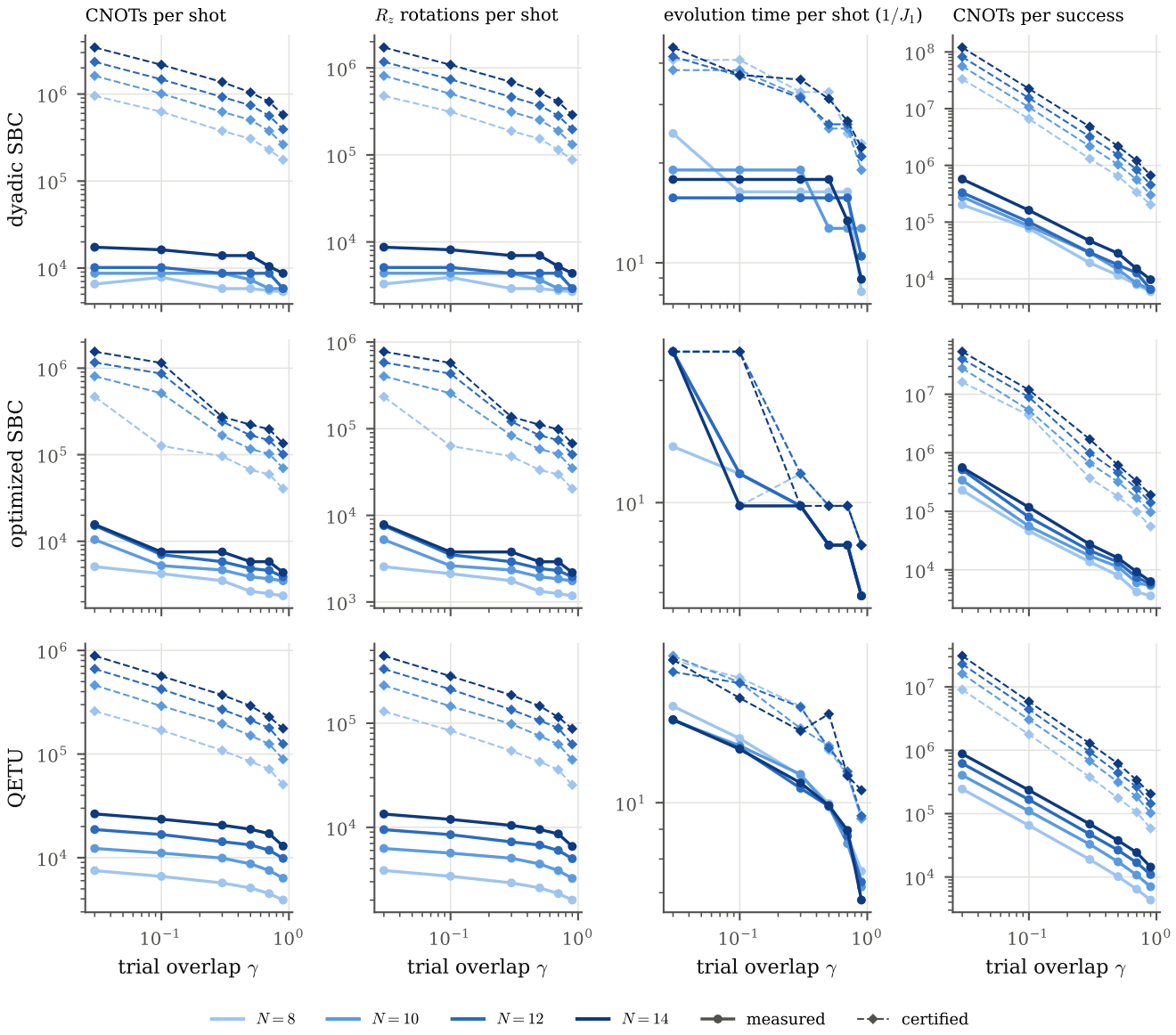


FIG. 4. Study A: filter-only resources at a gap held at $g = 0.40$ for every N, $\epsilon = 10^{-3}$, against the trial overlap. Rows: dyadic SBC, optimized SBC, QETU (Sec. VII). Solid: measured by simulation; dashed: certified. Columns 1–3 per shot, column 4 per success. Source: `results/gapov.jsonl`, `results/gapov_filter.json`.

and each factor answers to different inputs (Fig. 6).

Filter length. The physical evolution time depends on Δ and L only: fits give $T \propto \Delta^{-1.19} L^{0.77} N^{-0.24}$ measured and $T \propto \Delta^{-1.10} L^{0.84} N^{0.04}$ certified, so at fixed absolute gap the filter length is independent of N. The certified length is a median 2.0× the measured one (range 1.2–3): the certificate sizes the filter nearly right.

Trotter step. The measured dyadic filter tolerates coarse steps, median $\delta t = 1.25, 0.56, 0.30$ (in $1/J_1$) at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$: its pulses are long and smooth, and the filtered state lives where Trotter errors are small [19]. The certified step must cover the worst case over the whole evolution, median $\delta t = 0.017, 0.009, 0.005$, about 60–75 times finer and shrinking as T grows. The certified step count therefore grows as $L^{2.03}$ against $L^{0.71}$ measured and is a median 89.4× the measured one. Almost all of the certificate's overhead is in the Trotter bound, not in the filter.

Gates per step. One Strang step costs a median $20\,N$ CNOTs. The measured step count barely grows with N (exponent 0.08), so measured CNOTs grow as $N^{1.30}$; the certified step count grows as $N^{0.85}$ through the commutator constant and the scaled time, giving $N^{2.08}$.

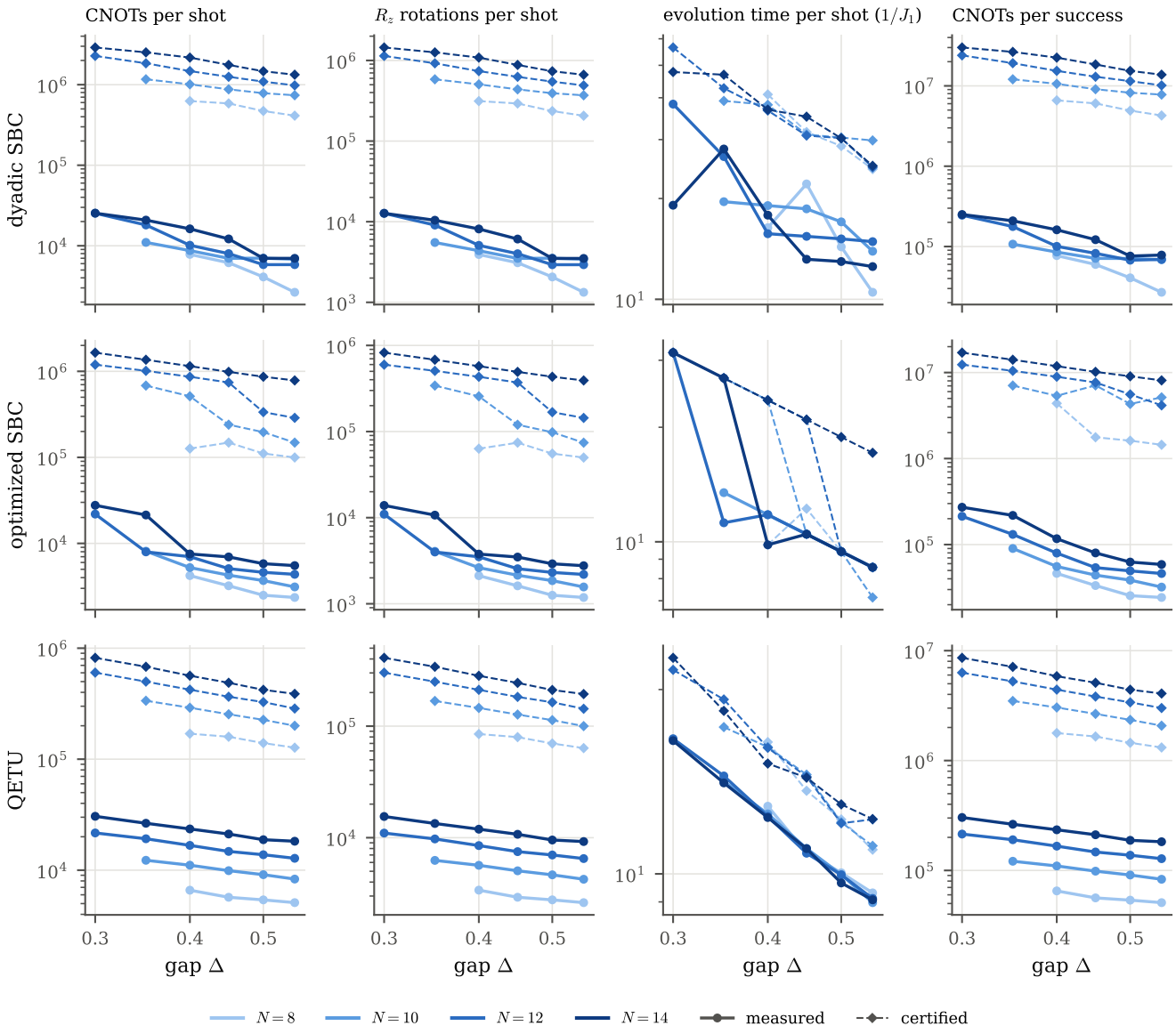


FIG. 5. Study B: as Fig. 4 at an overlap held at $\gamma = 0.10$ for every N, against the gap (held fixed across N by tuning J_2).

C. Cost laws

Fitting the expected CNOTs per success (filter and trial) to

$$C \approx a\, N^{\alpha}\, \Delta^{-\beta}\, L^{\lambda}\, \gamma^{-g}, \qquad L = \ln \frac{1-\gamma}{\gamma\epsilon}, \tag{12}$$

gives Table IV. For the dyadic filter,

$$C_{\mathrm{meas}} \approx 10.1\, N^{1.50} \Delta^{-1.15} L^{1.41} \gamma^{-0.78}, \tag{13}$$

$$C_{\mathrm{cert}} \approx 15\, N^{2.07} \Delta^{-1.27} L^{2.43} \gamma^{-0.98}, \tag{14}$$

within a typical factor of 1.23 and 1.22. The certified law has the $1/\gamma$ of the repetitions and an $L^{2.4}$ from the Trotter bound; the measured law grows more slowly in both N and L. Its γ exponent is below one because at coarse steps each Trotterized pulse moves some weight into or out of the ground state (p/γ ranges from 0.76 to 1.24). The certified cost lies 14–258× (median 62) above the measured one. These exponents summarize $N = 8$–14 and $g = 0.30$–0.55 and are not asymptotic laws. Appendix B compares the two SBC schedules and maps the cost over the (Δ, γ) plane.

V. CERTIFIED RESOURCES TO $N = 48$

Without simulation the pipeline needs only DMRG, so we plan for natural chains up to $N = 48$ with the $\chi = 2$ trial (Fig. 7, Table V). The second-order plan is 17–666×

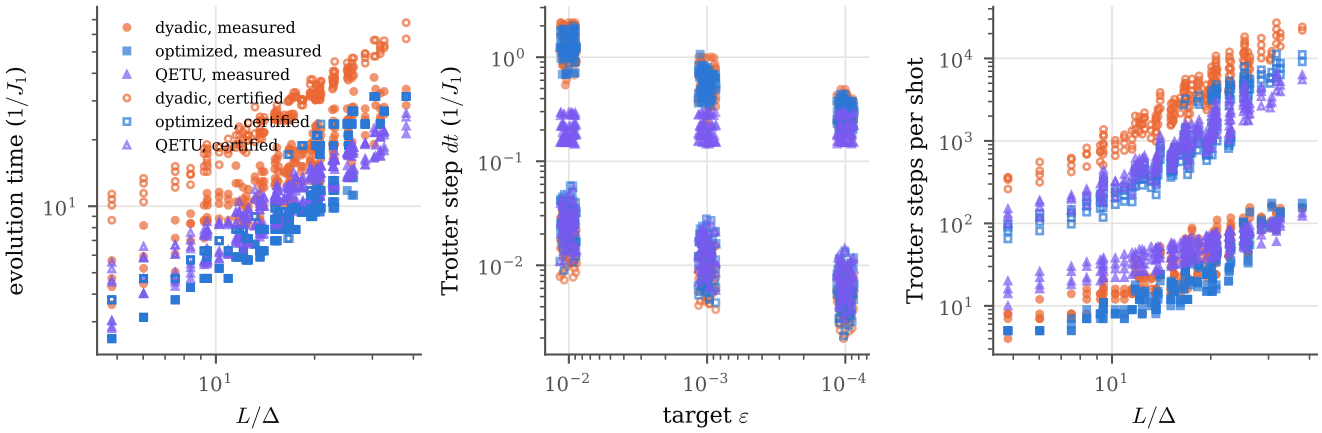


FIG. 6. All 222 cases. (a) Evolution time against L/Δ, $L = \ln[(1-\gamma)/(\gamma\epsilon)]$. (b) Trotter step against ϵ (jittered horizontally). (c) Trotter steps per shot against L/Δ. Filled: measured; open: certified. Orange circles: dyadic SBC; blue squares: optimized SBC; purple triangles: QETU.

TABLE IV. Fits of the expected CNOTs per success to Eq. (12) over all 222 gap/overlap cases. "rms"/"max": typical and worst factor between a case and its fit. Source: `results/gapov.jsonl`.

	cases	a	α	β	λ	g	rms	max
dyadic, measured	222	1.0×10^1	1.50	1.15	1.41	0.78	1.23	1.9
optimized, measured	222	9.8	1.50	1.36	1.24	0.83	1.30	2.3
QETU, measured	222	1.3×10^1	2.09	0.74	0.68	0.99	1.04	1.2
dyadic, certified	222	1.5×10^1	2.07	1.27	2.43	0.98	1.22	2.2
optimized, certified	222	1.9	2.27	1.30	2.74	0.97	1.45	3.2
QETU, certified	222	9.5	2.03	1.11	2.17	0.99	1.22	2.0

(median 93) cheaper than first order, and the optimized schedule 0.7–3.7× (median 2.9) cheaper than the dyadic one, but its design failed in 14 of 144 plans, all at $N \geq 32$, where the dyadic schedule remains feasible. For $N =$ 16–48 the expected CNOTs of the dyadic second-order plan grow as $N^{4.9}$, $N^{5.0}$ and $N^{3.5}$ for $J_2 = 0$, 0.2411, 0.4. These exponents combine the shrinking gap, the growing commutator constant and number of terms, and the decaying overlap. At $N = 48$, $J_2 = 0.2411$ the best plan (dyadic) needs 2.7×10^8 CNOTs per success.

A. What if the energy estimates are wrong?

The certificate is only as good as its inputs. We perturb them for $N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$ (Fig. 8) and, for the gap and overlap, for every $N \leq 14$, coupling and schedule (Table VI), and simulate the resulting circuits on the true trial state.

Gap. An assumed gap larger than the true one moves the stop-band edge above the first excitation. With exact inputs all 36 plans meet the target; a 10% overestimate breaks it in 2 (marginally, worst 1.1×10^{-2}), 25% in 10 and 50% in 16. The dyadic schedule is the most tolerant (0 misses at 25%) because its bound is loosest; the optimized and Chebyshev filters have no slack. Underestimating the gap is safe and cheap: assuming 0.5 of it costs 2.9×.

Ground energy and overlap. An error bar δE_0 lowers F_{pass}, and the cost stays flat until it is a sizeable fraction of the gap. Underestimating γ is safe. Overestimating it broke the target in 12 of 72 plans, all because the inflated γ crossed $1 - \epsilon$ and the pipeline dropped the filter; the empty filter should be accepted only on a certified lower bound on γ.

A margin from DMRG convergence. Setting δE_0 and δE_1 to the change between the last two bond dimensions of the DMRG ladder covered both exact energies in 75 of 75 tests on 15 systems with exact spectra, against 29 without error bars, and all 396 plans built from these ladders met the target. Once DMRG has converged the margin costs a median 1.00×; when it has not, the energy window becomes inconsistent and the pipeline reports that instead of a plan.

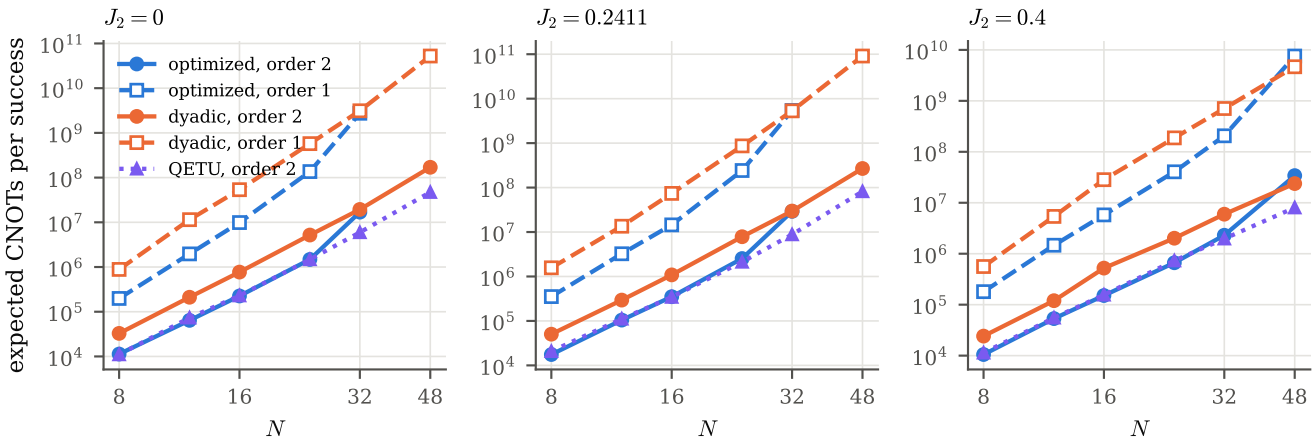


FIG. 7. Expected CNOTs per successful preparation versus N (trial $\chi = 2$, $\epsilon = 10^{-2}$), including the trial circuit and the repetitions (certified success probability). Solid: second order; dashed: first order. Missing optimized points: no feasible design (see text). Dotted triangles: QETU, second order. Sources: `results/scaling.json`, `results/ext_qetu_scaling.json`.

TABLE V. Cheapest second-order plan ("sch.": dyadic or optimized) for $\epsilon = 10^{-2}$, trial $\chi = 2$. Δ is the scaled gap, $T_{\text{phys}} = T/W$ in $1/J_1$. Steps for first (1st) and second (2nd) order with the same schedule; CNOT and T counts for second order per attempt (att.) and per success (succ.), divided by the certified success probability; "rot.": arbitrary rotations per attempt, synthesized to accuracy δ. Source: `results/scaling.json`.

N	J_2	γ	Δ	sch.	T_{phys}	p_g	steps (1st)	steps (2nd)	CNOT/att.	CNOT/succ.	rot.	δ	T/succ.
16	0	0.475	0.0206	opt.	14.3	0.38	38,020	464	7.8×10^4	2.3×10^5	3.9×10^4	9×10^{-8}	8.0×10^6
16	0.2411	0.703	0.0192	opt.	15.0	0.58	41,572	521	1.8×10^5	3.5×10^5	8.8×10^4	7×10^{-8}	1.3×10^7
16	0.4	0.911	0.0211	opt.	6.6	0.63	19,266	264	8.9×10^4	1.5×10^5	4.5×10^4	6×10^{-8}	5.4×10^6
24	0	0.328	0.0094	opt.	20.5	0.27	251,160	1,422	3.8×10^5	1.5×10^6	1.9×10^5	8×10^{-9}	6.0×10^7
24	0.2411	0.517	0.0087	opt.	16.3	0.29	248,256	1,329	7.0×10^5	2.5×10^6	3.5×10^5	5×10^{-9}	1.1×10^8
24	0.4	0.839	0.0102	opt.	10.8	0.59	87,533	700	3.7×10^5	6.6×10^5	1.9×10^5	1×10^{-8}	2.6×10^7
32	0	0.234	0.0054	opt.	80.0	0.23	3,103,919	9,847	3.5×10^6	1.7×10^7	1.8×10^6	1×10^{-9}	7.4×10^8
32	0.2411	0.373	0.0049	opt.	85.8	0.37	4,979,405	13,617	9.8×10^6	2.9×10^7	4.9×10^6	6×10^{-10}	1.3×10^9
32	0.4	0.768	0.0062	opt.	13.3	0.41	221,517	1,228	8.8×10^5	2.3×10^6	4.4×10^5	5×10^{-9}	9.6×10^7
48	0	0.125	0.0025	dya.	146.0	0.13	20,734,266	34,380	1.9×10^7	1.7×10^8	9.5×10^6	2×10^{-10}	8.2×10^9
48	0.2411	0.191	0.0022	dya.	142.5	0.19	27,888,279	41,586	4.6×10^7	2.7×10^8	2.3×10^7	1×10^{-10}	1.3×10^{10}
48	0.4	0.640	0.0031	dya.	69.3	0.64	4,816,485	12,414	1.4×10^7	2.4×10^7	6.9×10^6	6×10^{-10}	1.1×10^9

VI. BENCHMARK AGAINST THE MPS ALONE

The alternative to filtering a cheap trial is compiling a better MPS. We compare the two at equal target infidelity, measured on both sides: for each system the exact spectrum is computed per S^z block, every MPS trial with $\chi = 2, \ldots, 64$ is decomposed on it, and for each $\epsilon \in [10^{-5}, 10^{-2}]$ the cheapest route is found: the smallest MPS that meets $1 - \gamma \le \epsilon$ alone, against the cheapest smaller MPS followed by a measured SBC or QETU filter (shortest exact filter, then the coarsest passing Strang step), with cost (trial + filter)$/p$. Both routes use the staircase MPS cost of Sec. III.

Chains. On nine chains ($N = 12$–16, three couplings; Fig. 9) compiling the MPS is cheaper in most cases: the filtered route wins 18 of 63 targets, by 1.1–3.2×. Every win filters a trial that misses the target by $(1-\gamma)/\epsilon \le 8.8$, with a dyadic ladder of one to four pulses (1–21 Strang steps, $p \ge 0.988$), and all 90 candidate routes with a larger ratio lose, by a median 20×. The filter acts as a last-mile polisher: MPS cost jumps about fourfold per doubling of χ, so a target just below what a given χ reaches forces the MPS-alone route up a bond dimension, while a short filter removes the small residual of the lower one. The wins are therefore tied to the bond-dimension grid rather than to N or J_2. The optimized design never wins, because certifying suppression on the whole stop band needs $T \gtrsim \pi/\Delta$ where one dyadic pulse suffices,

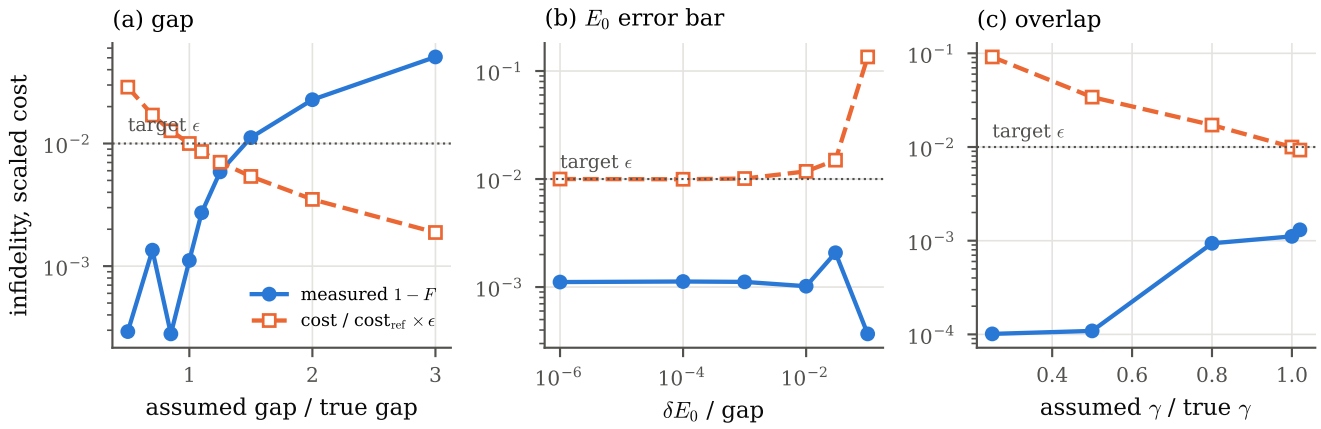


FIG. 8. Sensitivity to the inputs ($N = 12$, $J_2 = 0.2411$, $\epsilon = 10^{-2}$). Circles: measured infidelity of the planned circuit on the true trial state. Squares: expected CNOT cost relative to the plan with exact inputs, multiplied by ϵ to share the axis. (a) Assumed gap $f\times$ the true gap; (b) error bar on E_0; (c) assumed overlap $f\times$ the true γ. Source: `results/sensitivity.json`.

TABLE VI. Gap and overlap overestimation for every $N \in \{8, 10, 12, 14\}$ and J_2 (12 instances), $\epsilon = 10^{-2}$, second order: number of plans that miss the target and the worst measured infidelity. f: assumed over true gap or γ. Source: `results/ext_sensitivity.json`.

		dyadic		optimized		QETU	
input	f	miss	worst	miss	worst	miss	worst
gap	1	0/12	1.7×10^{-3}	0/12	4.2×10^{-3}	0/12	2.3×10^{-3}
gap	1.1	0/12	6.5×10^{-3}	1/12	1.1×10^{-2}	1/12	1.0×10^{-2}
gap	1.25	0/12	6.8×10^{-3}	5/12	2.6×10^{-2}	5/12	3.0×10^{-2}
gap	1.5	2/12	6.3×10^{-2}	7/12	6.8×10^{-2}	7/12	9.2×10^{-2}
gap	2	8/12	1.4×10^{-1}	7/12	1.5×10^{-1}	11/12	2.2×10^{-1}
gap	3	11/12	2.2×10^{-1}	12/12	2.5×10^{-1}	12/12	3.1×10^{-1}
γ	1.02	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}	1/12	2.4×10^{-2}
γ	1.05	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}	3/12	5.5×10^{-2}

and QETU beats the MPS alone at 11 targets but is never the cheapest route.

Cylinders. On 4×3 and 4×4 cylinders (periodic around the four-site circumference) at $J_2 = 0$ and 0.3, with snake-path MPS trials (Fig. 10), the picture reverses. A snake MPS struggles here: at 4×4, $J_2 = 0$, $\chi = 16$ misses $\epsilon = 10^{-2}$ ($1 - \gamma = 1.1 \times 10^{-2}$), $\chi = 32$ stalls at 7×10^{-4} and no $\chi \leq 64$ reaches 10^{-5}, while the compile cost grows about fourfold per doubling (5 328, 20 548 and 76 600 CNOTs for $\chi = 16$, 32, 64). The filtered route is cheaper at 27 of the 28 cylinder targets, by 1.1–8.8×, and is the only MPS-based route at 10^{-5} on 4×4. Every win uses the dyadic ladder, mostly on a $\chi = 16$ trial, with $p \geq 0.77$ and 1–46 Strang steps, and recovers factors $(1 - \gamma)/\epsilon$ up to 10^3. The filter's depth grows only as $\ln(1/\epsilon)$ at fixed gap, while the MPS needed to reach ϵ alone grows fourfold per bond-dimension step. All trials are snake MPS, a weak choice in two dimensions; a better ansatz would help both routes. The staircase cost charges every site a unitary on $b + 1$ qubits even where the bond is smaller, which favors the filtered route, most where its trial is cheap.

Certified. With certified filters the comparison is lost: the filtered route costs a median 152.3× the MPS alone on the chains (never below 13.5×) and a median 5.2× on the cylinders wherever some $\chi \leq 64$ reaches ϵ, winning only where no MPS does (Table VIII). The measured advantage on cylinders is a statement about what the circuit needs; claiming it with a guarantee requires the tighter Trotter bound discussed in Sec. VIII.

VII. COMPARISON WITH A CHEBYSHEV FILTER (QETU)

The pipeline also designs the quantum eigenvalue transformation of unitaries (QETU) [9]: d queries of $e^{-i\tau H_s \otimes Z_a}$ interleaved with ancilla rotations and a sin-

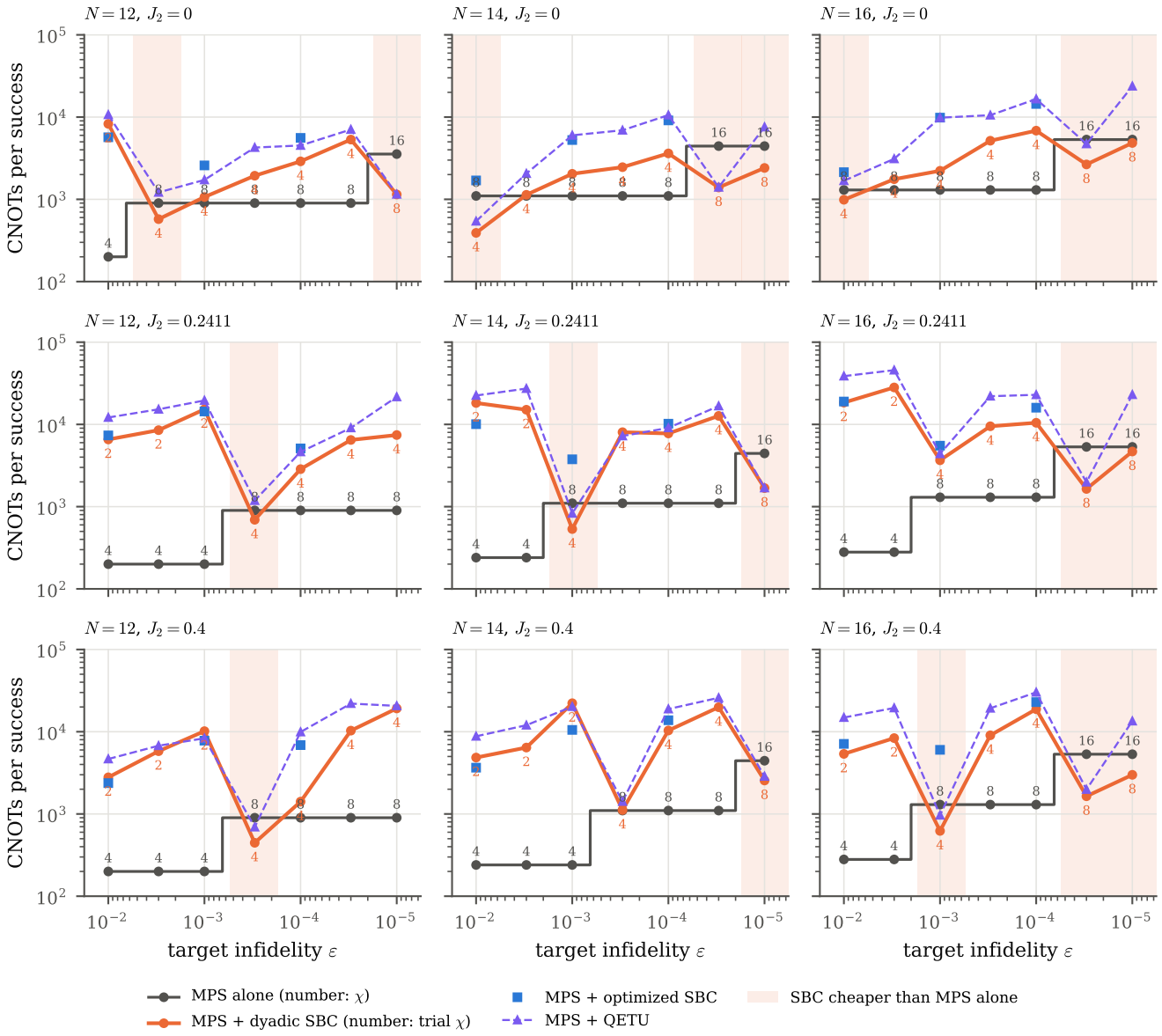


FIG. 9. Measured cheapest route to infidelity ϵ on chains of $N = 12$, 14, 16 (columns) at $J_2 = 0$, 0.2411, 0.4 (rows), all by exact simulation. Grey: cheapest MPS that meets $1 - \gamma \le \epsilon$ alone (number: χ). Orange: cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC (designed at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$ only); purple: QETU. Shaded: the SBC route is cheaper. Cost is (trial + filter CNOTs)/p. Sources: `results/emp_actual1d.jsonl` and, for $N = 16$, $J_2 = 0.2411$, `results/emp_actual.json`.

gle post-selection, whose ancilla-0 block is $p(\cos \tau H_s)$ for a real even polynomial p of degree d with $|p| \le 1$. With $\tau = \pi/2$ the stop band maps to $[0, x_s]$, $x_s = \cos(\pi\Delta/2)$, and we take the extremal Chebyshev filter

$$p(x) = \frac{T_d(x/x_s)}{T_d(1/x_s)}, \qquad \eta = \frac{1}{T_d(\cos(\tau w_0)/x_s)}, \tag{15}$$

certified in closed form, with $p(0) = 1$ and $T = d\tau \approx \Delta^{-1} \ln(2/\eta)$, two to three times less than the dyadic ladder. The Trotter bound, step allocation and Theorem 1 carry over unchanged; the classical phase-finding step is not costed, and QETU is costed in the controlled form (Sec. II).

Certified. QETU is feasible in all 36 second-order scaling cases and a median 3.3× (range 2.1–3.9) the cost of the dyadic schedule; against the optimized design the ratio is a median 1.00, the two being equal within the spread because both reach $T \approx \Delta^{-1} \ln(1/\eta)$ and the Trotter cost depends only on T. The certificate held in 24 of 24 simulated QETU plans, with a nearly tight leakage term (median 1.3×). In the controlled study the certified QETU plan is cheaper than the certified dyadic

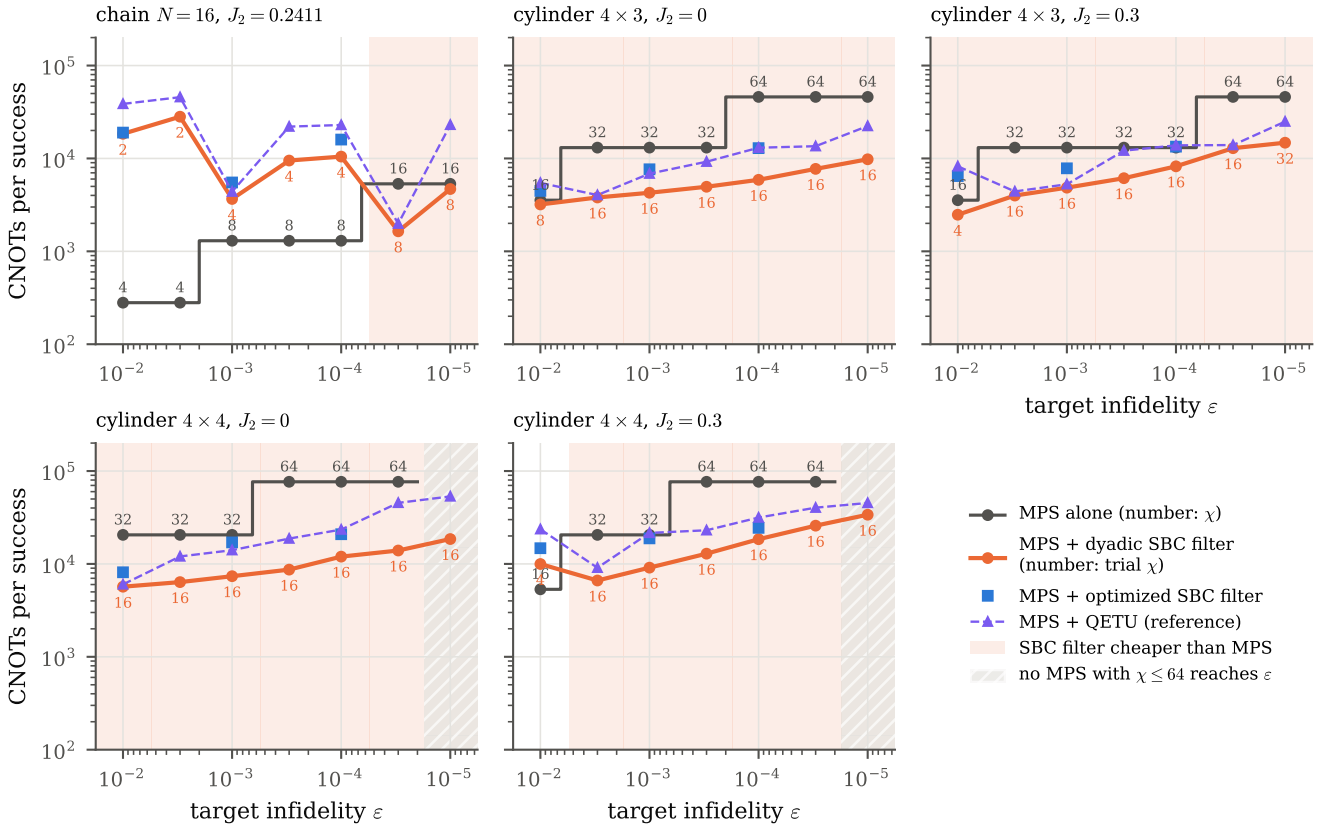


FIG. 10. Measured cheapest route to infidelity ϵ on the 16-site chain and on small cylinders. Grey: the cheapest MPS that meets the target alone (number: χ). Orange: the cheapest smaller MPS followed by the dyadic SBC filter (number: trial χ); blue squares: optimized SBC; purple: QETU. Shaded: the filtered route is cheaper; hatched: no MPS with $\chi \leq 64$ reaches ϵ. Every cost is (trial + filter CNOTs)/p from exact simulation. Source: `results/emp_actual.json`.

one in all 222 cases and the cheapest certified filter in 36, 49 and 74 of 74 at $\epsilon = 10^{-2}$, 10^{-3}, 10^{-4}. Certified, SBC offers no cost advantage over a polynomial filter; it avoids phase finding and measures after every pulse, which an early-abort protocol could exploit.

Measured. At $\tau = \pi/2$ QETU is cheaper in 0 of 74 cases at $\epsilon = 10^{-2}$, 9 of 74 at 10^{-3} and 56 of 74 at 10^{-4}. The reason is the Trotter step: QETU needs less evolution time (median 0.65× dyadic), but takes one fine Strang step per short query, $\delta t = \tau/W \propto 1/N$, so its step count is a median 1.40× the dyadic one and its CNOTs grow as $N^{2.22}$ against $N^{1.30}$. The long dyadic pulses tolerate steps of order $1/J_1$, but these shrink as roughly $\epsilon^{0.3}$.

Query length. Any $\tau \leq \pi/(1+\Delta)$ keeps the excited levels inside $|\cos \tau E| \leq \cos \tau\Delta$, so we redid every measured QETU search at $\tau = 3\pi/4$ and $\pi/(1+\Delta)$ (467 cases). One Strang step per query still suffices, so the cost falls to a median 0.68 of the $\tau = \pi/2$ cost at $\tau = \pi/(1+\Delta)$. With QETU at its best τ (Table VII) the SBC filters keep $\epsilon = 10^{-2}$ (7 QETU wins of 74), the two split 10^{-3} (38 of 74), and QETU wins 65 of 74 at 10^{-4}. Appendix C analyzes what sets the Chebyshev filter's degree.

TABLE VII. Cheapest measured filter in the controlled study (74 cases per target; dyadic/optimized/QETU) with QETU at $\tau = \pi/2$ and at its best query length $\tau \in \{\pi/2, 3\pi/4, \pi/(1+\Delta)\}$. Source: `results/qetu_tau.jsonl`.

target	$\tau = \pi/2$	best τ
$\epsilon = 1 \times 10^{-2}$	23/51/0	19/48/7
$\epsilon = 1 \times 10^{-3}$	16/57/1	4/32/38
$\epsilon = 1 \times 10^{-4}$	5/32/37	0/9/65

VIII. MEASURED VERSUS CERTIFIED

Every cost above has two versions (Table VIII). The measured cost picks the shortest filter and the coarsest Trotter step that reach ϵ, which needs the exact ground state; it shows what the circuit has to do. The certified cost is what the pipeline plans from Theorem 1 without that knowledge. They differ mainly through the Trotter step, and the difference decides which filter is cheapest: measured, the long SBC pulses win at loose targets; cer-

TABLE VIII. Each comparison measured (shortest filter and coarsest Trotter step that reach ϵ in exact simulation; uses the ground state) and certified (planned from Theorem 1; implementable). Controlled study: 74 cases per target. Crossovers: targets 10^{-2}–10^{-5}; certified SBC uses the pipeline's automatic choice of dyadic or optimized, QETU in parentheses. Sources: `results/gapov*.jsonl`, `results/emp_actual*.json*`, `results/cert_crossover.jsonl`.

comparison	measured	certified
Cheapest filter, $\epsilon = 1 \times 10^{-2}$ (dy/opt/QETU)	23/51/0	0/38/36
Cheapest filter, $\epsilon = 1 \times 10^{-3}$ (dy/opt/QETU)	16/57/1	0/25/49
Cheapest filter, $\epsilon = 1 \times 10^{-4}$ (dy/opt/QETU)	5/32/37	0/0/74
MPS + filter beats MPS alone, 9 chains	18 of 63	0 of 63 (QETU 0)
MPS + filter beats MPS alone, 4 cylinders	27 of 28	2 of 28 (QETU 3)

tified, every filter pays the same worst-case step per unit time, the shorter Chebyshev filter wins more often, and only the optimized SBC design competes.

To isolate the step, we keep the filter as planned and Trotterize it on a common grid $\delta t = 4 \cdot 2^{-n}/J_1$ for every chain ($N = 12$–16) and small-cylinder target and both schedules (182 cases; `run_aposteriori.py`). With the coarsest passing step the planned filter costs a median 0.09 (SBC) and 0.12 (QETU) of its certified cost (Table IX): the step, not the filter, is almost the whole overhead. A check that needs neither the bound nor the ground state, halving δt until the post-selected energy changes by less than $\epsilon\Delta/2$, met the target in 182 of 182 cases at a median 0.22 of the certified cost, but its calibration takes a median 2×10^{10} post-selected shots, growing as ϵ^{-2}. Tighter Trotter bounds, for example restricted to the low-energy subspace [19, 30], are what would close the gap; a better filter design would not.

IX. DISCUSSION

What sets the cost. The gap determines the coherent circuit: evolution time $\propto L/\Delta$, independent of N at fixed gap, and CNOTs and R_z per shot that scale as a power of $1/\Delta$. The overlap enters the circuit only through a logarithm and the expected cost through $1/\gamma$, so the filter tolerates a poor trial by paying in repetitions rather than depth; amplitude amplification [32, 33] would turn that into $1/\sqrt{\gamma}$ at the price of a deeper coherent circuit, and is not modeled. The target enters through L and, for SBC, through the Trotter step, $\delta t \propto \epsilon^{0.3}$.

Against the MPS alone. A filter pays where the trial is hard to improve. For chains the MPS infidelity falls quickly with χ, so the filter helps only as a last-mile polisher; on small cylinders, where a snake MPS needs a fast-growing bond dimension, it wins almost everywhere, measured. Certified, it does not, which again points to the Trotter bound as the place to improve.

Dyadic, optimized or Chebyshev. Measured, SBC wins at $\epsilon \approx 10^{-2}$ because its long smooth pulses tolerate coarse steps; QETU at its longest admissible query ties at 10^{-3} and wins from 10^{-4}, and certified it is usually the cheaper filter. The optimized SBC design is cheaper than the dyadic ladder at $\epsilon \leq 10^{-3}$ (median 0.89× measured, 0.35× certified) but occasionally fails; the dyadic ladder is the robust fallback, which is how the "auto" mode uses them.

What is guaranteed. Theorem 1 holds whenever its inputs are right. The Weyl bound and the filter certificate are rigorous, and the Trotter constants are exact up to the Lanczos tolerance. The interval for E_0, the bound on E_1 and the overlap come from DMRG and are well-converged estimates, not bounds; for $N \leq 14$ ED confirms them, and beyond that the certificate is conditional on all three. The gap is the critical input, and because its DMRG bias runs in the dangerous direction a deployable plan should carry an explicit margin δE_1, which costs little beside the Trotter overhead. Rigorous lower bounds on E_1 (e.g. Temple-type) and the Eckart inequality $1 - \gamma \leq (\langle H \rangle_\psi - E_0)/(E_1 - E_0)$ would remove these conditions.

Limitations. Noiseless simulation; all-to-all connectivity; repeat-until-success rather than early abort; trial circuits costed by a generic decomposition bound; exact simulation and the controlled study only to $N = 14$–16, sizes at which classical methods are exact, so the fitted exponents are descriptive and claims about larger systems rest on certified plans; and a fixable gap range (0.30–0.55) in the easy regime, well above the gaps near 0.07 of the natural chains at $N = 48$.

CODE AND DATA AVAILABILITY

The `sbpipe` package, the study scripts, all raw results (`sbpipe_study/results/*.json`) and the sources of this paper are available at https://github.com/aaronseymour7/FilterPipeline.

Appendix A: Proof of Theorem 1

Write the trial state as $|\psi\rangle = \sqrt{\gamma'}|g\rangle + \sqrt{1-\gamma'}|\chi\rangle$ with $|g\rangle$ in the ground space (energy $e_0 \in [0, w_0]$ in scaled units), $|\chi\rangle$ orthogonal to it and $\gamma' \geq \gamma$.

Lemma 1 (leakage)**.** *For the exact filter, $v = F(H_s)|\psi\rangle$ has weight outside the ground space at most ℓ, and $\|v\|^2 \geq p_g$.*

TABLE IX. Planned (Theorem 1) filters with an a-posteriori Trotter step. Columns: cases; median cost with the oracle step over the certified cost; energy-convergence rule (targets met, median cost over certified, median post-selected shots for the calibration). Source: `results/aposteriori.jsonl`.

		oracle		energy convergence	
schedule, system	cases	cost/cert.	met	cost/cert.	shots
SBC, chain, $N = 12$	21	0.08	21/21	0.22	2×10^{10}
SBC, chain, $N = 14$	21	0.07	21/21	0.22	9×10^{10}
SBC, chain, $N = 16$	21	0.08	21/21	0.19	3×10^{11}
SBC, 4×3	14	0.19	14/14	0.32	1×10^{11}
SBC, 4×4	14	0.14	14/14	0.23	1×10^{11}
QETU, chain, $N = 12$	21	0.10	21/21	0.22	3×10^{9}
QETU, chain, $N = 14$	21	0.10	21/21	0.22	4×10^{9}
QETU, chain, $N = 16$	21	0.10	21/21	0.21	6×10^{9}
QETU, 4×3	14	0.21	14/14	0.34	6×10^{9}
QETU, 4×4	14	0.15	14/14	0.24	1×10^{10}

Proof. $F(H_s)$ is a function of H_s, so $v = \sqrt{\gamma'}F(e_0)|g\rangle + \sqrt{1-\gamma'}F(H_s)|\chi\rangle$ with the two parts orthogonal. Every component of $|\chi\rangle$ has energy in $[\Delta, 1]$, so $\|F(H_s)\chi\| \le \sup_{[\Delta,1]}|F| \le \eta F_{\text{pass}} \le \eta|F(e_0)|$. The excited weight is $x/(a+x)$ with $a = \gamma' F(e_0)^2$, $x = (1-\gamma')\|F\chi\|^2 \le (1-\gamma')\eta^2 F(e_0)^2$; it is increasing in x and, after dividing by $F(e_0)^2$, decreasing in γ', which gives ℓ. Finally $\|v\|^2 \ge a \ge \gamma F_{\text{pass}}^2$. □

For pure states the trace distance is $\sin\angle$, so $D(\hat{v}, |g\rangle) = \sqrt{\ell'} \le \sqrt{\ell}$ ($\hat{v} = v/\|v\|$).

Lemma 2 (implementation error)**.** *Let* $B_i = \frac{1}{2}(e^{-i\phi_i}\tilde{S}_i(t_i) + e^{i\phi_i}\tilde{S}_i(-t_i))$ *be the ancilla-0 block of the implemented pulse and* $C_i = \cos(H_s t_i + \phi_i)$*. Then* $\|\prod_i B_i - \prod_i C_i\| \le \epsilon_U$.

Proof. $C_i = \frac{1}{2}(e^{-i\phi_i}e^{-it_iH_s} + e^{i\phi_i}e^{it_iH_s})$, so $\|B_i - C_i\| \le \frac{1}{2}(\|\tilde{S}_i(t_i) - e^{-it_iH_s}\| + \|\tilde{S}_i(-t_i) - e^{it_iH_s}\|) \le \epsilon_i$, where ϵ_i is the product-formula bound (symmetric in the sign of t) plus the synthesis error of the pulse's rotations. Since $\|B_i\|, \|C_i\| \le 1$, telescoping gives $\|\prod B_i - \prod C_i\| \le \sum_i \epsilon_i$. A synthesized trial circuit prepares $|\tilde{\psi}\rangle$ with $\|\tilde{\psi} - \psi\| \le R_{\text{trial}}\delta$, and $\|\prod B_i\| \le 1$, so the total error of the unnormalized output is at most ϵ_U. □

With $\tilde{v} = \prod B_i|\tilde{\psi}\rangle$, $\|\tilde{v} - v\| \le \epsilon_U$. The distance from v to the line through $\tilde{v}$ is at most $\|v - \tilde{v}\|$, so $\sin\angle(v, \tilde{v}) \le \epsilon_U/\|v\| \le \epsilon_U/\sqrt{p_g}$. The triangle inequality for the trace distance gives the bound on D. The states $\hat{v}$, $|g\rangle$ and the normalized output are pure, so $D^2 = 1 - |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$; since $|g\rangle$ lies in the ground space, $\text{Tr}(\Pi_0\tilde{\rho}) \ge |\langle g|\tilde{\psi}_{\text{out}}\rangle|^2$, which gives (8). (For a mixed output, e.g. under noise, a trace-distance bound alone would not bound the ground-space weight by D^2; this case is outside the theorem.) The success probability is $\|\tilde{v}\|^2 \ge (\|v\| - \epsilon_U)^2 \ge (\sqrt{p_g} - \epsilon_U)^2$. For the energy, the normalized state has weight at least $1 - D^2$ on the ground space and at most D^2 on levels $\le E_{\text{top}}$, so $\langle H\rangle - E_0 \le (E_{\text{top}} - E_0)D^2$. □

Appendix B: Dyadic versus optimized SBC, and cost maps

The two SBC schedules reach the same physics by different routes. The dyadic ladder has zero phases, so it passes the ground state with $f(E_0) = 1$ and its length grows in whole pulses. The optimized design (times and phases from SBC's recipe with a certified suppression constraint) is shorter, at the price of $f(E_0) < 1$: its measured success probability is a median 0.88 times γ (as low as 0.52). Measured per shot, the optimized filter uses a median 0.58 times the dyadic evolution time and 0.62 times its Trotter steps; its tolerated step is the same, median $\delta t = 1.24$ at $\epsilon = 10^{-2}$ and 0.30 at 10^{-4}. Per success, after the lower success probability is paid, it is still a median 0.89× the dyadic cost (range 0.58–1.55) and the cheaper one in 169 of 222 cases. Certified, the advantage is larger, a median 0.35× (cheaper in 221 of 222), because the optimized filter's leakage bound is nearly tight while the closed-form dyadic bound is loose (Sec. III A), and the shorter filter needs a looser Trotter budget; its certified cost lies 5–185× (median 26) above the measured one, against 62× for dyadic.

Counting the cheapest of the three filters case by case, the optimized SBC filter wins 51 of 74 cases at $\epsilon = 10^{-2}$ (dyadic the other 23), 57 at 10^{-3} (dyadic 16, QETU 1), and 32 at 10^{-4}, where QETU takes 37 and dyadic 5. The cost laws of Table IV make the trade-off explicit: the optimized measured cost grows more steeply with the gap ($\Delta^{-1.36}$ against $\Delta^{-1.15}$), because its design is matched to the gap, and less steeply with L. The price of the optimized schedule is its design: a nonconvex optimization of about 40 s per case that converged to a filter with

$f(E_0)^2 \sim 10^{-3}$ in 21 of 286 measured points and had to be repeated with other seeds; the dyadic ladder needs no design and never fails.

Cost maps.

Two maps make the interplay visible. At fixed N and $\epsilon = 10^{-2}$ (Fig. 11), the contours of the dyadic cost per success run nearly parallel to the gap axis for $g \gtrsim 0.40$: there the overlap alone sets the cost, from 2–6 $\times 10^3$ CNOTs at $\gamma = 0.9$ to 1–4 $\times 10^5$ at $\gamma = 0.03$; below $g \approx 0.40$ the gap starts to matter, and $g = 0.30$ costs 1.3–2.7 times $g = 0.40$ at the same overlap. The optimized filter is cheaper over most of the map, and its contours tilt toward the gap axis over the whole range, as its steeper gap exponent implies. Per shot (not shown) the contours of both turn by ninety degrees and the gap alone sets the circuit. At fixed gap and overlap (Fig. 12), the measured cost per success grows 7–21× (dyadic) from $\epsilon = 10^{-2}$ to 10^{-5} and 1.8–4.6× from $N = 8$ to 14; the contours tilt toward the ϵ axis, so over this range the target matters more than the size. The optimized filter's lead is largest at loose targets and at $\gamma = 0.5$, and shrinks toward $\epsilon = 10^{-5}$.

Appendix C: What the Chebyshev filter needs

To see how the gap and the overlap act on the Chebyshev filter itself, we replace the certificate by simulation for $N = 8$–16 (675 cases). Each trial state is decomposed on the exact eigenbasis (dense diagonalization per S^z block), so the smallest QETU degree $d_{\min}$ that reaches ϵ for that state is found exactly; at that degree we then find the smallest number of Strang steps per query with the Trotterized circuit simulated on the full state vector, allowing a slightly higher degree when it makes one step per query enough. Three families separate the inputs. In the first, $J_2 = 0.2411$ for every N and the gap closes with N. In the second, J_2 is chosen at each N (by bisection on ED) so that the gap is $E_1 - E_0 = 0.40$ exactly ($J_2 = 0.384, 0.510, 0.539, 0.550, 0.555$ for $N = 8$–16). In the third, J_2 is tuned at $N = 10$, 12 and 14 to gaps from 0.30 to 0.55. For the trial states we use the optimal MPS at $\chi = 1$, 2, 4 and synthetic states $\sqrt{\gamma}|E_0\rangle + \sqrt{1-\gamma}|\xi\rangle$ with γ from 0.03 to 0.9, ξ being the normalized excited part of the $\chi = 2$ MPS, so that the overlap changes while the shape of the excited weight does not. $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$.

The degree is set by the gap and the target alone. The measured degrees collapse onto

$$d_{\min}\,\Delta \;\simeq\; \frac{2}{\pi}\cosh^{-1}\frac{1}{\eta}, \qquad \eta^2 = \frac{\gamma\epsilon}{1-\gamma}, \tag{C1}$$

for every family, trial and target (Fig. 13b). This is the certified degree with the true gap: the certificate asks for a median 1.09 times $d_{\min}$. Where the excited weight of the trial sits does not help. The Chebyshev filter is equiripple, so every state above the gap is suppressed by the same factor $1/T_d(1/x_s)$, and a trial whose excited weight lies far above the gap needs the same degree as one whose weight sits at the gap. Over the gap scan $d_{\min} \propto \Delta^{-1.00}$.

The certificate's Trotter count is pessimistic, not its filter. One Strang step per query was enough in all 675 cases, against a certified median of 16 steps per query. Commutator bounds are worst-case over all states, but the filter only has to act well on a state supported near the bottom of the spectrum. As a result the certified cost is a median 22 times what the filter needs (range 7–2 $\times 10^2$), and the real cost follows the degree:

$$\mathrm{cost}_{\mathrm{real}} \;\approx\; \frac{c_{\mathrm{step}}(N)}{\gamma}\,d_{\min} \;\propto\; \frac{c_{\mathrm{step}}(N)\,W}{g\,\gamma}\,\ln\frac{1-\gamma}{\gamma\epsilon}, \tag{C2}$$

with c_{step} the CNOTs of one step ($\propto N$) and $g = E_1 - E_0$. Fitting the simulations gives overlap $\gamma^{-0.97}$ once the logarithm is divided out ($\gamma^{-1.21}$ as a plain power, against $\gamma^{-1.45}$ for the certified plan) and gap $\Delta^{-0.98}$ (certified $\Delta^{-1.38}$). With the gap held at 0.40 the cost grows as $N^{2.3}$; at fixed J_2, where g also closes, as $N^{3.1}$ (Fig. 13c). The difference is the $1/g$ of Eq. (C2).

Gap and overlap are not independent inputs. Tuning J_2 to hold the gap moves the chain toward the dimerized point $J_2 = 1/2$, where a $\chi = 2$ MPS is close to exact: at the gap-matched couplings the $\chi = 2$ overlap is 0.97–1.00 at every N, against 0.70–0.90 at $J_2 = 0.2411$. Up to $J_2 \approx 1/2$ a larger gap and a better trial come together, so for real trials the two effects compound; beyond it the gap keeps growing while the $\chi = 2$ overlap falls again (to 0.78 at $J_2 = 0.68$, $N = 14$). Holding one of them fixed through J_2 therefore always moves the other, which is why the scans here use synthetic trials and Sec. IV a sublattice rotation of a real trial circuit.

Appendix D: Reproducing the results

The code, raw results and paper sources are at `https://github.com/aaronseymour7/FilterPipeline`. After `pip install -r requirements.txt`, all commands below run from the repository root. `python sbpipe_study/run_study.py all` runs the DMRG front end (cached), the validation (A), scaling (B), sensitivity (C), ansatz (D) and oracle (E) studies and writes `results/*.json`; `python sbpipe_study/make_figures.py` writes every figure, table and the numerical macros used in this text. The stored results were produced with per-term CNOT counts and converted to the bond compile of Eq. (9) by `sbpipe_study/recompile_results.py`, which checks every stored count against the per-term formula before replacing it; a fresh run counts bonds directly. `python -m pytest sbpipe/tests` runs the component checks (Pauli algebra, Weyl bound, Trotter bounds against dense matrix exponentials, the Dirichlet identity (3), filter certificates against a dense grid, gate counts against an explicit merged sequence and against qiskit, and an

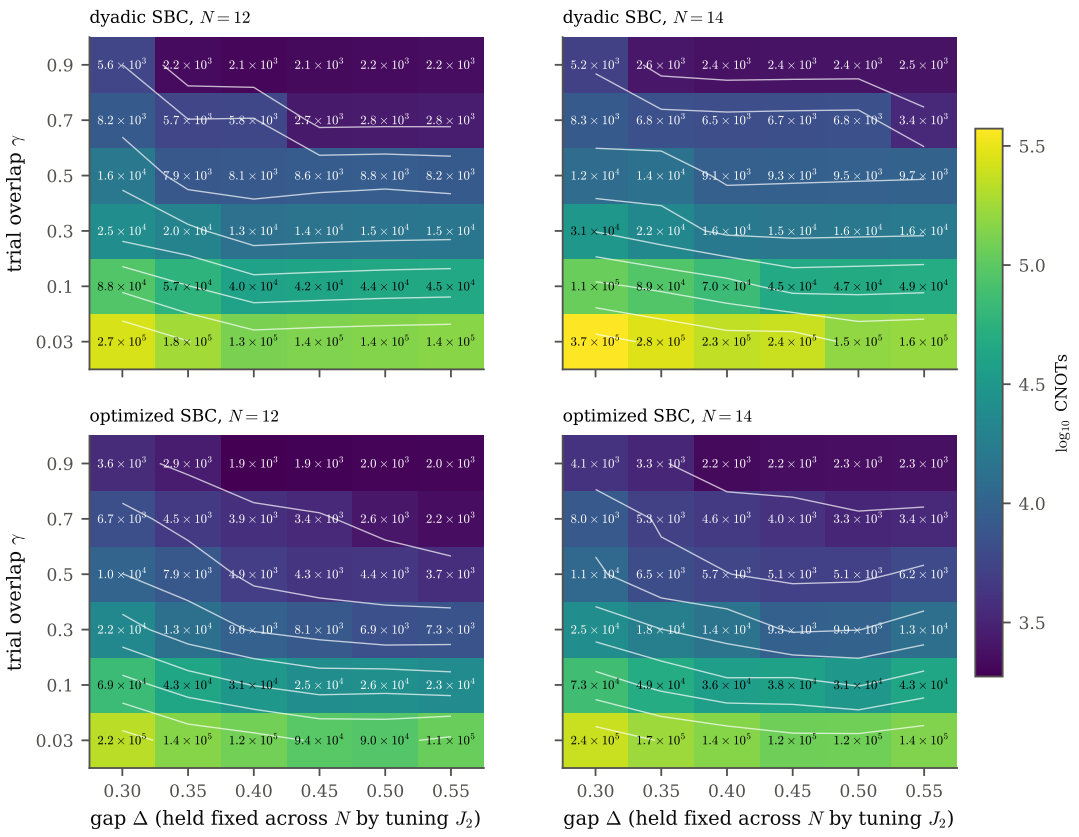


FIG. 11. Filter CNOTs per success over gap and overlap at $\epsilon = 10^{-2}$, measured, for the dyadic (top) and optimized (bottom) SBC filters at $N = 12$ and 14. White contours every factor 1.8. Source: `results/gapov.jsonl`, `results/gapov_grid.jsonl`.

end-to-end certificate check).

[1] I. Stetcu, A. Baroni, and J. Carlson, Projection algorithm for state preparation on quantum computers, Phys. Rev. C **108**, L031306 (2023), arXiv:2211.10545.

[2] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of Trotter error with commutator scaling, Phys. Rev. X **11**, 011020 (2021).

[3] S. R. White, Density matrix formulation for quantum renormalization groups, Phys. Rev. Lett. **69**, 2863 (1992).

[4] U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, Ann. Phys. **326**, 96 (2011).

[5] S.-J. Ran, Encoding of matrix product states into quantum circuits of one- and two-qubit gates, Phys. Rev. A **101**, 032310 (2020).

[6] Y. Ge, J. Tura, and J. I. Cirac, Faster ground state preparation and high-precision ground energy estimation with fewer qubits, J. Math. Phys. **60**, 022202 (2019).

[7] L. Lin and Y. Tong, Near-optimal ground state preparation, Quantum **4**, 372 (2020).

[8] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, Quantum singular value transformation and beyond, in *Proc. STOC 2019*, p. 193.

[9] Y. Dong, L. Lin, and Y. Tong, Ground-state preparation and energy estimation on early fault-tolerant quantum computers via quantum eigenvalue transformation of unitary matrices, PRX Quantum **3**, 040305 (2022).

[10] M. Bee-Lindgren, Z. Qian, M. DeCross, N. C. Brown, C. N. Gilbreth, J. Watkins, X. Zhang, and D. Lee, Controlled gate networks: theory and application to eigenvalue estimation, Eur. Phys. J. A (2025), doi:10.1140/epja/s10050-025-01750-y, arXiv:2208.13557.

[11] S. Fomichev, K. Hejazi, M. S. Zini, M. Kiser, J. Fraxanet, P. A. M. Casares, A. Delgado, J. Huh, A.-C. Voigt, J. E. Mueller, and J. M. Arrazola, Initial state preparation for quantum chemistry on quantum computers, PRX Quantum **5**, 040339 (2024).

[12] D. W. Berry, Y. Tong, T. Khattar, A. White, T. I. Kim, G. H. Low, S. Boixo, Z. Ding, L. Lin, S. Lee, G. K.-L. Chan, R. Babbush, and N. C. Rubin, Rapid initial-

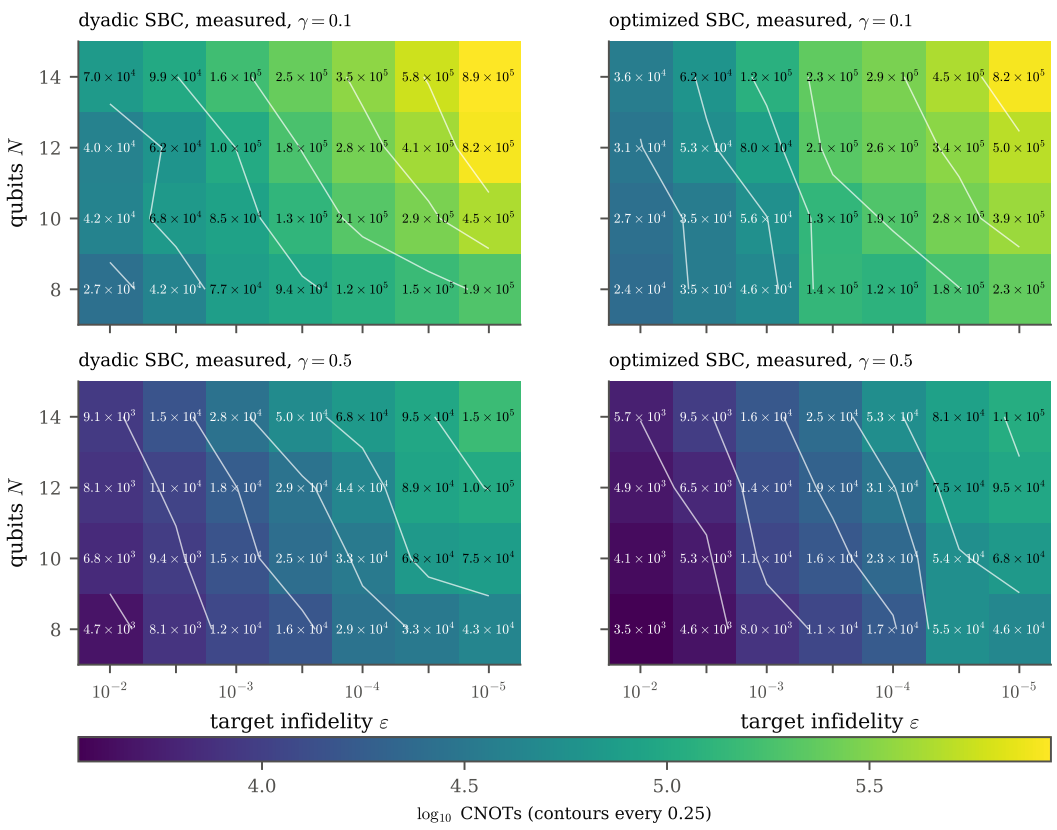


FIG. 12. Filter CNOTs per success over N and ϵ at a gap held at $g = 0.40$, measured, for the dyadic (left) and optimized (right) SBC filters at $\gamma = 0.1$ (top) and 0.5 (bottom). Source: `results/gapov*.jsonl`.

state preparation for the quantum simulation of strongly correlated molecules, PRX Quantum **6**, 020327 (2025).

[13] K. Choi, D. Lee, J. Bonitati, Z. Qian, and J. Watkins, Rodeo algorithm for quantum computing, Phys. Rev. Lett. **127**, 040505 (2021).

[14] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+T approximation of z-rotations, Quantum Inf. Comput. **16**, 901 (2016).

[15] V. V. Shende, S. S. Bullock, and I. L. Markov, Synthesis of quantum-logic circuits, IEEE Trans. Comput.-Aided Des. **25**, 1000 (2006).

[16] K. Okamoto and K. Nomura, Fluid-dimer critical point in $S = 1/2$ antiferromagnetic Heisenberg chain with next nearest neighbor interactions, Phys. Lett. A **169**, 433 (1992).

[17] S. Eggert, Numerical evidence for multiplicative logarithmic corrections from marginal operators, Phys. Rev. B **54**, R9612 (1996).

[18] J. Gray, quimb: a python package for quantum information and many-body calculations, J. Open Source Softw. **3**, 819 (2018).

[19] B. Şahinoğlu and R. D. Somma, Hamiltonian simulation in the low-energy subspace, npj Quantum Inf. **7**, 119 (2021).

[20] Z. Qian, J. Watkins, G. Given, J. Bonitati, K. Choi, and D. Lee, Demonstration of the rodeo algorithm on a quantum computer, arXiv:2110.07747.

[21] E. Rule, I. Stetcu, and J. Carlson, Simplified projection on total spin zero for state preparation on quantum computers, arXiv:2410.02848 (2024).

[22] T. D. Cohen and H. Oh, Optimizing rodeo projection, arXiv:2305.19952 (2023).

[23] M. Patkowski, O. Ayyildiz, K. Hunt, N. Jansen, and D. Lee, Improved rodeo algorithm performance for spectral functions and state preparation, arXiv:2602.05978 (2026).

[24] C. F. Kane, N. Gomes, and M. Kreshchuk, Nearly-optimal state preparation for quantum simulations of lattice gauge theories, arXiv:2310.13757 (2023).

[25] T. R. Müller, M. Geiger, and C. B. Mendl, Ground-state preparation of the Fermi-Hubbard model on a quantum computer with 2D topology via quantum eigenvalue transformation of unitary matrices, arXiv:2411.18535 (2024).

[26] E. Karacan, Y. Chen, and C. B. Mendl, Enhancing scalability of quantum eigenvalue transformation of unitary matrices for ground state preparation through adaptive finer filtering, arXiv:2401.09091.

[27] G. Lee, M. Kang, J. Hong, S. Fomichev, and J. Huh, Filtered quantum phase estimation, arXiv:2510.04294 (2025).

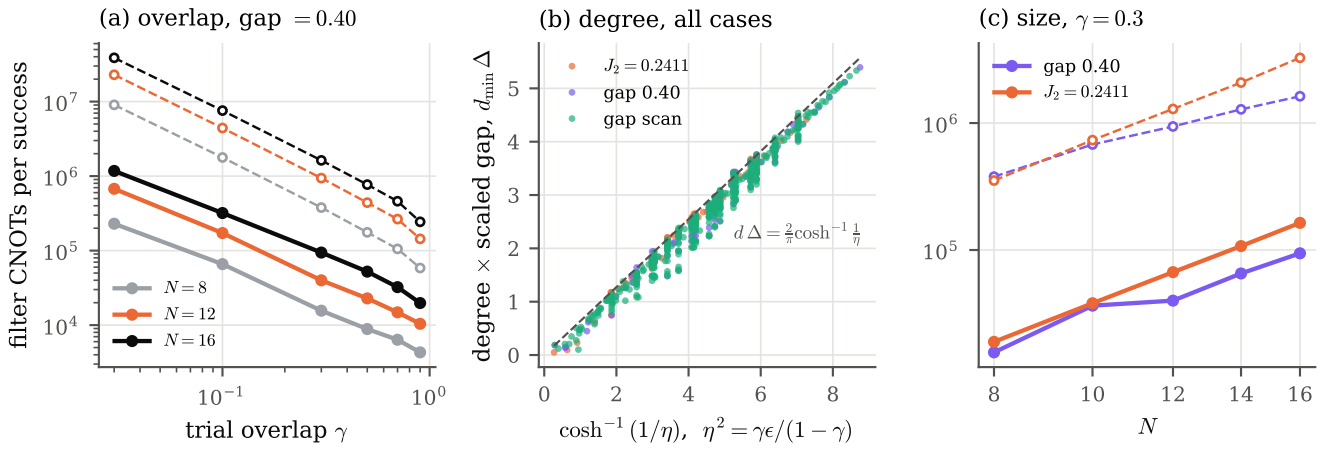


FIG. 13. What the filter actually needs, from exact spectra and state-vector simulation ($N = 8$–16, QETU, second order). (a) CNOTs per success versus overlap at a gap held at 0.40, $\epsilon = 10^{-3}$: simulated (filled) and certified (open, dashed). (b) Smallest degree times scaled gap for all 675 cases against $\cosh^{-1}(1/\eta)$; dashed: Eq. (C1). (c) Cost versus N at $\gamma = 0.3$, $\epsilon = 10^{-3}$, with the gap held at 0.40 or J_2 held at 0.2411. Source: `results/emp_trotter.json`, `sbpipe_study/run_empirical.py`.

[28] M. Watanabe, H. Nishi, T. Kosugi, S. Tsuneyuki, and Y. Matsushita, Overcoming the matrix-product-state encoding barrier via DMRG-guided probabilistic imaginary-time evolution, arXiv:2605.30141 (2026).

[29] A. M. Childs, D. Maslov, Y. Nam, N. J. Ross, and Y. Su, Toward the first quantum simulation with quantum speedup, Proc. Natl. Acad. Sci. **115**, 9456 (2018).

[30] K. Mizuta and T. Kuwahara, Trotterization is substantially efficient for low-energy states, arXiv:2504.20746 (2025).

[31] M. Amy, D. Maslov, and M. Mosca, Polynomial-time T-depth optimization of Clifford+T circuits via matroid partitioning, IEEE Trans. Comput.-Aided Des. **33**, 1476 (2014).

[32] G. Brassard, P. Høyer, M. Mosca, and A. Tapp, Quantum amplitude amplification and estimation, Contemp. Math. **305**, 53 (2002).

[33] T. J. Yoder, G. H. Low, and I. L. Chuang, Fixed-point quantum search with an optimal number of queries, Phys. Rev. Lett. **113**, 210501 (2014).